Serial access memories

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Outline

- **Introduction**
- **Magnetic disks**
- **Solid State Units**
- **Magnetic tape**
- **Optical disks.**

Introduction

Serial access memories (disks, tapes, cassettes) are generally used for secondary and off-line memory.

Secondary memory is usually composed of magnetic disks and flash memories.

Optical disks, cartridges and tapes normally form the off-line memory.

Serial access memories are characterized by low cost and high access time.

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Magnetic disks

The memory element is a disk covered with magnetic material on which there are a number of concentric tracks.

The memory unit can be made up of multiple disks: in this case they are connected to a single axis and rotate at constant speed.

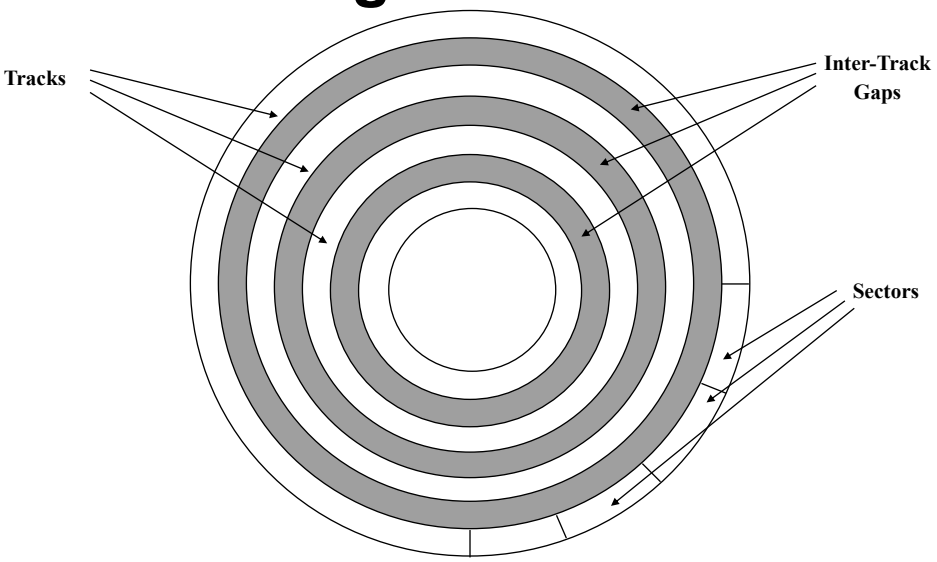
Each surface (*face*) has a head able to move radially to the desired track.

The various heads usually move in a solid manner.

The set of tracks at equal distance from the center placed on different faces is called a *cylinder*.

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Organization



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Organization (II)

All tracks contain the same number of bits but have a different length.

The linear density of information thus increases towards the center.

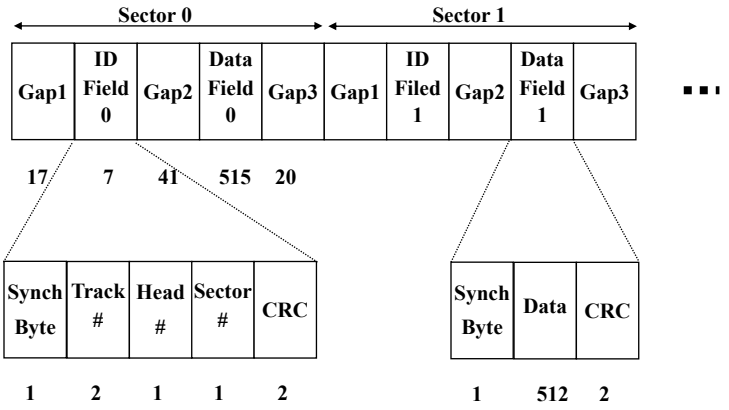
Each track is organized in sectors, corresponding to the transfer unit.

Each sector is subdivided into two fields, separated by special gap fields:

- **the *identifier* field**
- **the *data* field.**

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Example: Seagate ST506 hard disk



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Disk access

When you want to read / write a data, you need to

- **place the head over the track, moving the head**
- **place the sector under the head by moving the disk**
- **read/write the sector.**

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Access time

The access time t_A is composed of 3 parts:

- **t_S: time to position the head on the correct track (*seek time*)**
- **t_L: time to position the disk so that the correct sector is under the head (*latency time*)**
- **t_D: time to serially read the data (*data-transfer time*).**

Hence

$$t_A = t_S + t_L + t_D$$

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Block access

Since t_S are t_L are comparable with t_D, data are normally grouped in *sectors* (whose typical size is about 1 KB), and the read / write operations use the sector as the minimal data transfer unit.

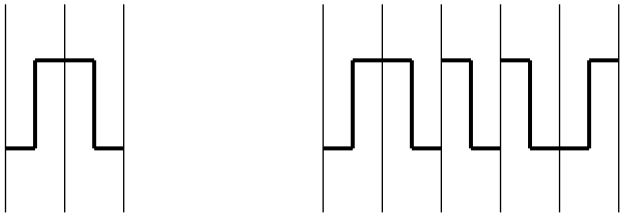
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Data encoding

It is introduced to avoid using a clock track for synchronization.

A very simple encoding technique is the so called *phase encoding* or *Manchester*.

Each bit of information is stored as a rising or falling transition.



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Typical parameters

- **Capacity: hundreds of GBs to some TBs**
- **Form factor: 8'', 5,25'', 3,5'', 2,5'', 1,8'', 1'', 0,85'' (the first two are now obsolete)**
- **Seek time: from 5 to 15 ms**
- **Transfer rate: from 40 to 130 MiB/s**
- **Rotation speed: from 5,400 to 15,000 rpm.**

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Amazon.com (5-5-2019)

Best Seller



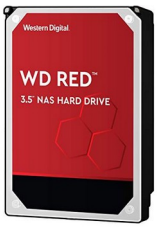
Western Digital WD Blue 1TB PC Hard Drive - 7200 RPM Class, SATA 6 Gb/s, 64 MB Cache, 3.5" - WD10EZEX

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Western… …de 1TB Hard D… …RPM Class, SATA 6 Gb/… …MB Cache, 3.… WD10EZEX

★★★★½ ~ 10,316

$49.99 ~~$109.99~~

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SATA revision 3.0 (6 Gbit/s, 600 MB/s, Serial ATA-600) [edit]

Serial ATA International Organization (SATA-IO) presented the draft specification of SATA 6 Gbit/s physical layer in July 2008,[19] and ratified its physical layer specification on August 18, 2008.[20] The full 3.0 standard was released on May 27, 2009.[21]

Third-generation SATA interfaces run with a native transfer rate of 6.0 Gbit/s; taking 8b/10b encoding into account, the maximum uncoded transfer rate is 4.8 Gbit/s (600 MB/s). The theoretical burst throughput of SATA 6.0 Gbit/s is double that of SATA revision 2.0. It is backward compatible with SATA 3 Gbit/s and SATA 1.5 Gbit/s.[19]

The SATA 3.0 specification contains the following changes:

- 6 Gbit/s for scalable performance.
- Continued compatibility with SAS, including SAS 6 Gbit/s, as per "a SAS domain may support attachment to and control of unmodified SATA devices connected directly into the SAS domain using the Serial ATA Tunneled Protocol (STP)" from the SATA Revision 3.0 Gold specification.
- Isochronous Native Command Queuing (NCQ) streaming command to enable isochronous quality of service data transfers for streaming digital content applications.
- …n NCQ management feature that helps optimize performance by enabling host processing … management of outstanding NCQ commands.

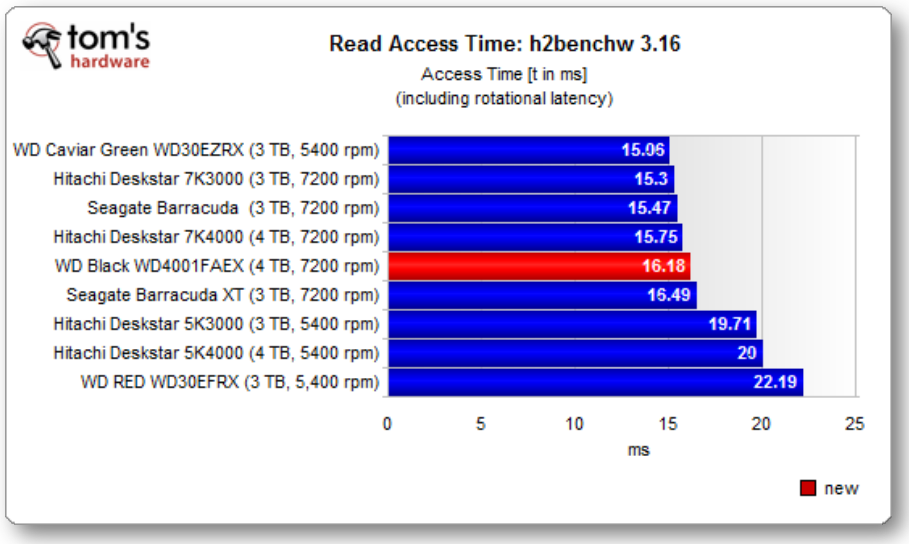


…7200RPM 64MB …6.0Gb/s …rage) 3.5" Internal … Year Warranty

…ck - order soon.

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Winchester Hard Disks

They take the name from the IBM internal code of the first product of this type, the 3340 model disk, introduced in 1973.

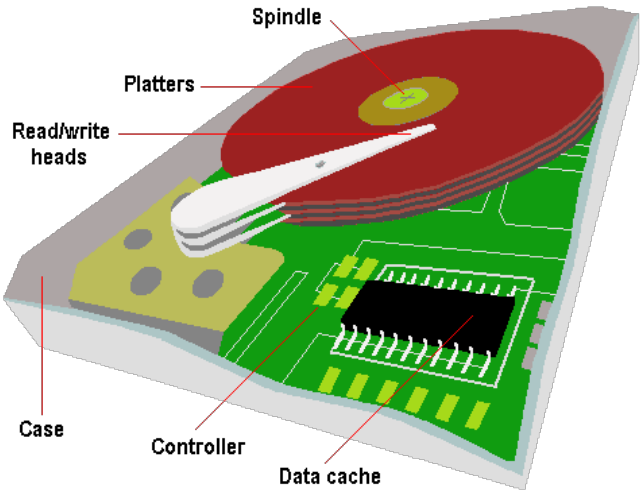
In this case, the head is very close to the disk surface, so that it allows higher storage density.

In practice, it rests on the disc when it is stationary, and is raised by this by an aerodynamic effect produced by the "air" cushion generated by the rotation.

The disc and the head are sealed in a special casing.

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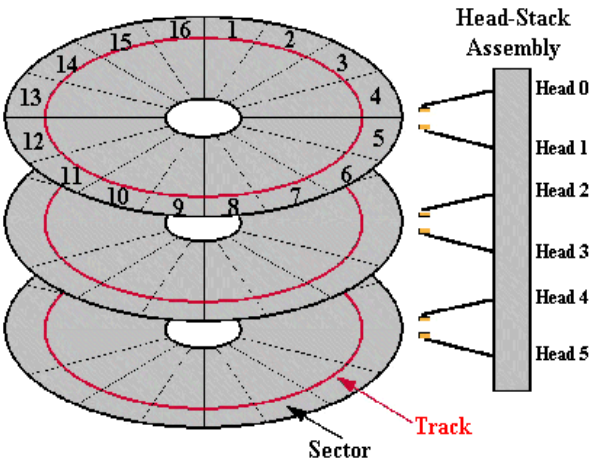
Winchester Hard Disk



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Winchester Hard Disk

Drive Physical and Logical Organization



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Disk driver and disk controller

The disk driver connects a disk with the disk controller.

It can include

- **a data buffer, consisting of a semiconductor memory, which can play the role of disk cache**
- **an IDE, SATA, SCSI or other interface.**

The disk controller interfaces with the processor and memory.

Data transfer between the disk controller and the main memory is often done in DMA.

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Solid state units

They are NOT serial access memories but are increasingly used in the secondary memory.

They are made with electronic technologies.

They have no moving parts.

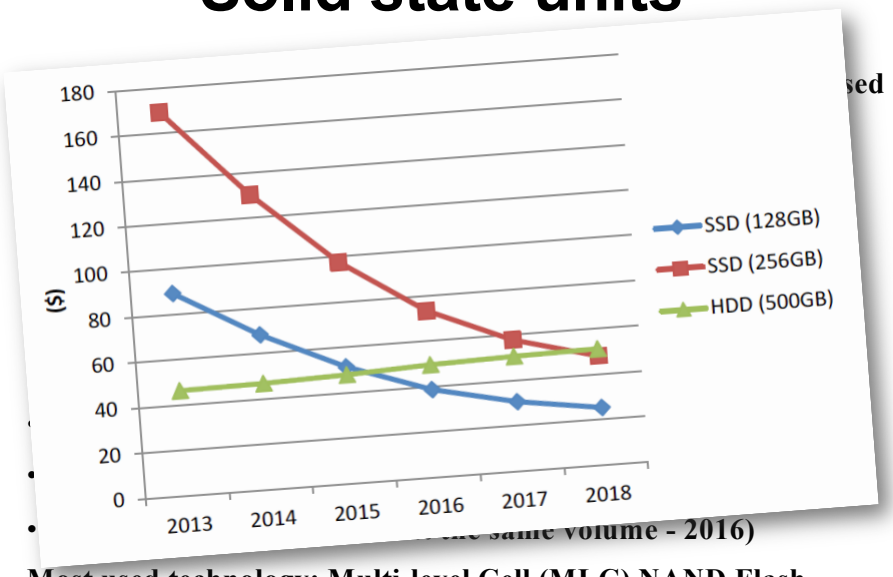
Comparison with traditional hard disks:

- **More resistant to physical shocks**
- **Minor access time (random access)**
- **Less latency**
- **Less consumption**
- **More expensive (about 4x at the same volume - 2016)**

Most used technology: Multi-level Cell (MLC) NAND Flash.

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Solid state units



- sed
- ume - 2016)

Most used technology: Multi-level Cell (MLC) NAND Flash.

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Solid state units

Key features (consumer devices)

- **Capacity up to 1TB**
- **Transfer rates up to ~ 500MB / s**
- **Limited number of rewriting operations (2000 ÷ 3000)**
- **Necessary "Wear leveling"**
- **Problem of "Write amplification"**
- **Guaranteed operation for more than 10 years for "typical" use on a PC (~ 10GB of data per day).**

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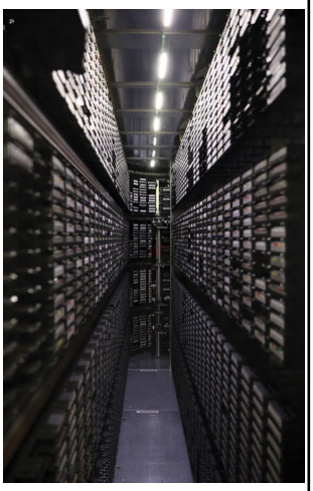
Off-line memories

In some contexts, there is a need to store large amounts of digital data (such as movies)

In these cases, access time is not a problem

The important parameters are

- **The cost, including the maintenance cost (which depends on the obsolescence of media and formats adopted)**
- **Duration**
- **The footprint**
- **For these applications the following solutions are used:**
  - **Magnetic tapes**
  - **Cartridges**
  - **Optical Memories**



IEEE Spectrum, May 2017

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Magnetic tapes

In this case, the support is a 1/4 or 1/2 inch wide flexible plastic tape on which 9 parallel tracks are stored.

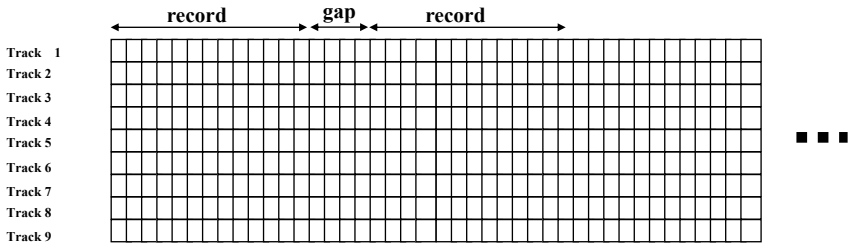
Each track has its own head, so it is possible to read / write the 9 tracks at the same time.

The bits of each byte are distributed among the first 8 tracks; the last one contains a parity code.

The data is organized into *records*, which can be of fixed size (and thus replaceable by others) or variable (optimizing the use of the tape).

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Tape organization



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Cartridges

Compared to tapes, cartridges have the following differences:

- **the tape width is 8 mm**
- **the read / write system is analogous to that of tape (helical scan in the direction of width).**

The storage capacity is between 2 and 5 Gbyte.

The transfer rate is around some hundred Kbyte / sec.

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Magnetic tapes applications

They represent the most common means of off-line storage, typically to handle backups in the data center.

With the spread of automatic tape handling devices (called *robots*), it has also enabled them to be used as online memories in applications where acceptable access times are a few seconds / minute.

In 2014, Sony developed a disk drive with a capacity of 185 TB.

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Optical memories

They are diffused in the form of optical disks.

They have the following characteristics:

- capacity around the GiByte
- high access times (1s)
- data transfer rate as in hard-disks (tens of Mbyte / s)
- high reliability (due to the lack of close or contact mechanical parts, such as magnetic disks and tapes).

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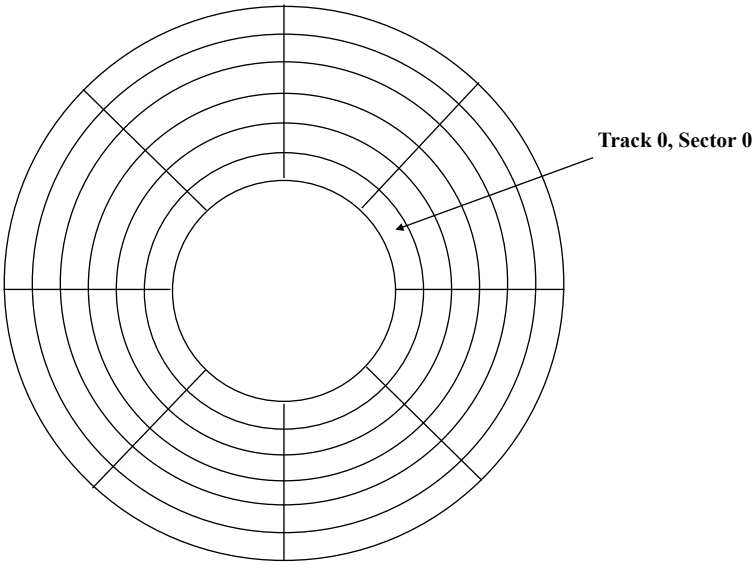
Organization

Two different solutions can be adopted:

- CAV (*Constant Angular Velocity*)
  - it is the solution adopted on magnetic disks
  - the disk is organized in concentric tracks, and the outer tracks have less linear storage density
  - each sector is accessible directly by specifying track and sector number
- CLV (*Constant Linear Velocity*)
  - this is the solution for music CDs, CD-ROMs and DVDs
  - the laser beam marks the ditches / pitches at constant linear velocity (with variable angular velocity), following a spiral trajectory
  - at the beginning of each sector, an identifier specifies the current value of the minute / second / block; casual access is more laborious.

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CAV



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CLV

Typical values for a CR-ROM:

- **Track distance: 1.6 μm**
- **Radius: 32.55 mm**
- **Number of spires: 20.344**
- **Total length of the track: 5.27 Km**
- **Speed: 1.2 m/sec**
- **Time to go through the whole track: 4,391 sec = 73.2 min**
- **Density: 176.4 Kbyte/sec**
- **Total capacity: 774,57 Mbyte.**

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Optical disk types

- **CD**
- **CD-ROM**
- **WORM**
- **CD-RW**
- **DVD**

They differ for

- **Capacity**
- **Alterability**
- **Mechanism of operation.**

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